

SAI BONTHA

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EXPERIENCE

Embedded Software Engineer

Jan 2026 – Present

Dell Technologies

Austin, TX

- Develop and maintain **multithreaded embedded C/C++ applications (POSIX threads/mutexes/semaphores)** on **2 Embedded Linux platforms** for system management, device monitoring, and peripheral integration over **UART/SPI/I2C/TCP/IP**
- Support **device-driver integration**, **device-tree** configuration, **BSP** modifications, and **Linux startup**
- Debug system-level failures layer by layer — reproduce the issue, inspect **serial console/system logs**, trace execution with **GDB**, audit **driver/device-tree/interface** settings to isolate the fault; validate fixes with hardware and QA and **regression-test**, cutting average investigation time for comparable issues by **20%**
- Automate firmware deployment, system configuration, and log collection with **5 Bash scripts**, reducing repetitive manual work and improving consistency
- Support **firmware and system validation** across **25 test scenarios** (device initialization, peripheral communication, software integration, **regression testing**); investigated **15 system-level defects**

Embedded Software Engineer

Jun 2021 – Nov 2023

HCLTech

Hyderabad, India

- Developed **8 C/C++ modules** for peripheral communication and system integration on **2 ARM-based Embedded Linux**
- Implemented **multithreaded applications** and **IPC (POSIX threads, shared memory, message queues, sockets)**
- Supported **device-driver integration**, **kernel/U-Boot** configuration, **device-tree** updates, **root filesystem** customization, **cross-compilation**, and **firmware image** builds
- Debugged application and hardware integration issues with **GDB**, Linux logs, and serial tools — personally fixed or investigated **30 defects**; supported **40** unit, integration, and regression test scenarios
- Reproduced defects with hardware and QA teams, investigated **root causes**, and validated fixes for **6 firmware releases**

PROJECTS

Linux Character Device Driver | C, Linux Kernel

github.com/saihontha18-hub/linux-char-driver

- Implemented a **character device driver** with **alloc_chrdev_region/cdev_add**, per-minor message buffers with **mutex** protection, and an **ioctl interface** (reset/get/set/stats) shared with user space via a common header
- Wrote an **end-to-end user-space test harness** exercising open/read/write/ioctl across multiple device minors

UART Data Logger | Python

github.com/saihontha18-hub/uart-data-logger

- Built a **CLI tool** logging serial frames to **timestamped CSV** with **automatic reconnect** on port failure for unattended dev-board test runs
- Supports **line mode** and **fixed-frame mode** with sync-byte framing and **XOR checksum** validation, plus live throughput stats

CAN Frame Codec | C11, CAN Bus

github.com/saihontha18-hub/can-frame-codec

- CAN 2.0A/B frame codec** in strict C11: **integer-only bit-timing solver** (exhaustive BRP/TSEG search) and **1741-check suite** (DLC 0–8 round-trips, arbitration); clean under **-Wall -Wextra -Werror -pedantic**

OPEN SOURCE

Zephyr RTOS | C, I2S

github.com/zephyrproject-rtos/zephyr/pull/121168

- Upstream pull request implementing **config_get** for the **LiteX and MAX32 I2S drivers**, returning stored stream configuration with host-side tests (approved by maintainers)

scpi-parser | C, Test Instrumentation

github.com/j123b567/scpi-parser/pull/173

- Upstream pull request: **SCPI_NO_HIDDEN_VISIBILITY** opt-out fixing **-Werror** build breaks on bare-metal toolchains (open)

littlefs | C, Embedded Filesystem

github.com/littlefs-project/littlefs/pull/1233

- Upstream **bug fix**: getslice hid the directory entry after a pending move; verified with the reporter's **regression test** (under review)

EDUCATION

Youngstown State University

Youngstown, OH

M.S. Electrical and Computer Engineering

Jan 2024 – Dec 2025

TECHNICAL SKILLS

Languages: C, C++, Python, Bash

Embedded: Embedded Linux, Bare-Metal, Device Drivers, BSP, Board Bring-Up, U-Boot, Device Tree, Yocto/Buildroot, FreeRTOS, Zephyr, ARM, Microcontrollers

Protocols: UART, SPI, I2C, CAN, GPIO, TCP/IP

Tools: GDB, JTAG, Oscilloscopes, Logic Analyzers, Git, CI/CD